

Simple Solar Daylight Lamp Based On Solar Energy Harnessing For Apartments

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Rooftop solar photovoltaic (PV) installations are gaining popularity. Usually these are grid connected systems. Their power levels range from one kilowatt to a few tens of kilowatts. These are being installed on the roofs of small buildings, office buildings, individual houses, and bungalows, etc. Even in apartment complexes, on the roofs of the apartment buildings big solar systems are being installed. The power generated by this system is used for common area lighting, lifts, etc, in that apartment complex (net metering).

However, for individual apartments of high rise buildings, there are no PV installations. As the number of apartments in cities are increasing, we have to look for ways and means to install PV panels there. This will open up new opportunities for harnessing solar energy.

Challenges

Before any system is proposed for the apartments, there is a need to identify the issues and concerns. Following are some of the main reasons apartments do not have PV installations:

Safety. Solar panels are quite heavy and use glass panes. Hence, if installed in the high rise buildings, there are chances that these may fall down due to various reasons. This poses threat to life. Unless it is completely safe, we cannot install solar panels in high rise buildings.

No space. It is very difficult to find a suitable place where panels could be installed. The only possibility is on the balcony wall or grills.

Less sunlight. Most apartments get sunlight only for four to five hours. If the balcony is east-facing, sunlight is available in the morning and for west-facing balcony in the afternoon. The amount of power generated is

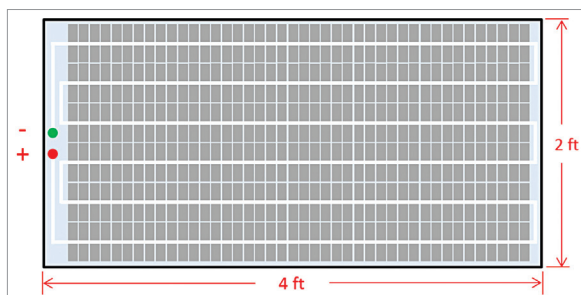


Fig. 1: Polymer panel Type I, suitable for horizontal mounting

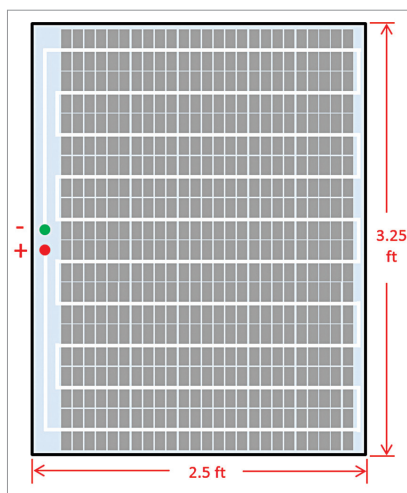


Fig. 2: Polymer panel Type II, suitable for vertical mounting

less. Therefore, it is not an attractive proposition for apartment owners.

Rented apartments. For the apartments given on rent, the tenant may be interested in having solar power, but the owner may not permit it, because of so many issues. Therefore, the system should be such the tenants should be able to quickly install the panels. Whenever required, the removal also should be easy.

In spite of above problems, it is worth exploring the possibility of harnessing solar energy in apartments. Otherwise, a big section of population will be deprived of solar energy.

It is worth making a humble beginning. As the technology advances, it may become more efficient and cost-effective. Here is an attempt to design a PV system for this segment.

Standardisation

In order to make effective use of limited space and limited power generation potential, it becomes necessary to standardise various aspects of the design. A detailed discussion on each of the design parameters to be standardised is given below.

PV voltage. PV panels are available in various voltage ratings. For roof-top applications with sufficient space, there is no restriction on the number of panels and operating voltage. However, in the proposed design, there may be space to install only a few small panels. Hence, there is a need for standardising the PV voltage. It will encourage people to design appliances for a standard voltage and will drive down the cost.

Further, due to limited space, the amount of power generated will be in the range of 100 to a few hundred watts. At these power levels, it is not possible to design a grid connected system. It has to be an off-grid system. Therefore, extra wiring is required in the house. From wiring point of view, with thinner cables, wiring becomes cheaper. Thin wires can be easily concealed without spoiling the interiors of the apartment.

Considering above points, and taking into account the safety of the user, we can standardise the PV voltage to 120V DC. Some countries have 120 volts as standard, and it may offer some advantages too (to be explored). A PV panel with 36 cells